Take-Home Assignment for Segment 10

Write a program to find best matches to a protein query in a protein database (same task as blastp but using a much simpler method). The sequence similarity between proteins X and Y is measured by Jaccard index (https://en.wikipedia.org/wiki/Jaccard_index) of the tetrapeptide composition, and is defined as the number of tetramers that are found in both proteins X and Y, divided by the number of tetramers that are found in at least one of the proteins X and Y.

Input:

Two FASTA files with protein sequences:

1. Query sequence (a single sequence in the file)
2. Database to be searched, typically sequences of all proteins in a genome

Output:

Top 5 hits with the highest similarity of tetramer composition to the query sequence (using Jaccard index as a measure of similarity)

Maximum reward for completing the assignment is 20 points.

Notes and recommendations:

- I recommend using a 4D array of type bool or char, such as Counts[21][21][21][21], to store info about tetramers that are present or absent in a protein. If you convert the amino acid sequence to a sequence of numbers 0-20 (the 20 amino acids plus X for any/unknown) you can set all values in the array to false and scan the sequence to set the values for tetramers present in the protein to true.
- You need two such arrays: one for the query and one for the protein you just read from the database. Once you calculate the similarity of the protein to the query, you can reuse that array for the next protein.
- You can create the array dynamically if you prefer, but they are not too large for the stack as long as you do not try to make one for each protein in the database, which I would not recommend anyway.
- You can use parts of the code from the THA06 assignment (finding the most hydrophobic proteins).
- **Do not use the <algorithm> container (#include <algorithm>) and do not use any external function to calculate the Jaccard index** – it is not too hard to write your own.
- **If anything is unclear, don't be afraid to ask.**

Submit your code via eLC.